

AutoGarage Project Module Documentation & Status Report

Date: August 18, 2026 | **Project:** AutoGarage Workshop Management | **Tech Stack:** Django, Python, HTML5/CSS3, SQLite

1. Overview of Completed Modules & Role Dashboards

A. Owner / Admin Dashboard & Staff Management

- **Owner KPI Dashboard:** Dedicated executive dashboard (*/owner/*) featuring KPI summary widgets for total monthly revenue, active jobs count, completed jobs count, total registered customers, total vehicles, pending unpaid invoices count, staff count, and low-stock alert badges.
- **Revenue & Staff Performance Analytics:** Dynamic revenue breakdown charts, mechanic productivity summary tables (tracking total completed jobs & labor revenue per mechanic), recent job cards log, and recent invoices audit trail.
- **Staff & Role Customization:** Create, edit, and delete staff accounts (Advisor, Mechanic, Store Manager, Custom Role), assign user privileges, set custom role display titles (e.g., Senior Diagnostician), and enforce role-based access control (RBAC).
- **Incentives & Expense Calculator:** Calculates net workshop profit (Revenue - Overhead Expenses - Parts Used) and individual mechanic commission allocations.
- **Garage Settings & Workshop Branding:** Dedicated admin settings page (*/settings/*) to configure Garage Name, Tagline, Phone, Email, Address, GSTIN, and Logo upload for official receipts.

B. Service Advisor Operations & Customer CRM

- **Advisor Operations Dashboard:** Dedicated dashboard (*/advisor/*) showing assigned active jobs, pending vehicle check-ins, customer service queue, and quick actions for rapid Job Card creation.
- **Customer CRM & Service History:** Complete customer directory, contact info, and linked vehicle repair history.
- **Vehicle Registry & Instant Search API:** Registry for Make, Model, Year, Plate, VIN, Mileage, Photo upload, and an instant AJAX search lookup API (*/api/search-records/*) to auto-fill customer and vehicle history when creating job cards.
- **Job Card Management & Workflow:** Automated job numbering (JC-00001), status workflow (Pending → In Progress → Waiting for Parts → Completed → Delivered), labor cost entry, progress photo uploads with captions, and automatic Title Case text formatting.
- **Invoicing & Billing System:** Converts completed Job Cards into tax invoices with automatic computation of spare parts totals, labor charges, pickup/drop fees, 5% GST, AMC discounts, partial payments, and remaining balance due.

C. Store Manager & Inventory Control

- **Store Manager Inventory Dashboard:** Specialized inventory dashboard (*/store/*) displaying low stock alert banners, total inventory valuation, category breakdown, supplier activity, and pending parts requests.
- **Spare Parts Catalog:** Comprehensive stock tracking with part numbers, unit prices, minimum stock warning levels, categories, and supplier references.
- **Stock Transactions & Audit:** Logs Stock In and Stock Out operations with reference numbers and notes.
- **Inventory Export:** One-click export of parts list to **CSV** and **PDF** report formats.

D. Mechanic Workbench & My Jobs Dashboard

- **Mechanic My Jobs Dashboard:** Dedicated, clean workbench dashboard (*/mechanic/*) displaying assigned repair jobs sorted by status (Pending, In Progress, Waiting for Parts).
 - **Job Detail & Repair Logs:** View repair instructions, problem descriptions, assigned vehicle photos, allocated spare parts list, and update repair progress status in real-time.
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2. Modules Currently In Progress

A. WhatsApp Notification Workflow

- Integrated WhatsApp dispatch view (/send-whatsapp/) to send job completion alerts, invoice billing summaries, and AMC reminders with audit logging.

3. What is Pending & Open Requirements

A. Pending: Annual Maintenance Contract (AMC) Module Integration

Key Pending Challenge:
The basic AMC database models (*AMCPlan*, *CustomerAMC*, *AMCServiceSchedule*) and rudimentary views have been created. However, **full business logic integration remains pending due to implementation complexity.**

Specific Challenges / Missing Logic for AMC:

1. **Job Card Integration:** Auto-detecting active AMC contracts during Job Card creation to apply free service vouchers.
2. **Service Schedule Tracking:** Auto-generating quarterly service visits upon AMC purchase and linking scheduled visits to Job Cards.
3. **Invoice Discount Engine:** Enforcing AMC discount rules (e.g., 100% off labor, 10% off spare parts) dynamically on billing.
4. **Renewal Automation:** Automated background reminders for contracts expiring within 30 days.

B. Additional Set of Pending Requirements

1. **Mobile Responsiveness & Handheld UI:** Full mobile design (collapsible sidebar, responsive table stacks, mobile-friendly forms) is currently pending.
2. **Role-Specific Dashboard UI Polish:** Custom dashboard widgets for store managers and mechanics.
3. **Print-Ready Documents:** Clean A4 print stylesheets for Invoices and Job Cards.

4. Issues, Blockers & Delays

Issue / Blocker	Category	Description	Mitigation Plan
AMC Workflow Uncertainty	Feature Architecture	Lack of technical design linking CustomerAMC <-> JobCard <-> Invoice.	Define a step-by-step AMC logic plan & discount rule matrix.
Mobile Responsiveness	UI / UX Layout	Complex tables and multi-column forms stretch on small mobile screens.	Add CSS media queries, card views for tables, and responsive mobile nav.
Staff Incentive Edge Cases	Business Logic	Uncertainty in profit sharing formula per mechanic vs flat labor %.	Define commission formula (flat labor % vs net profit allocation).
WhatsApp Gateway Setup	Third-Party API	Templates use wa.me links rather than official direct API endpoints.	Decide between client-side wa.me or official Twilio/Meta API.
Stock Deduction Safety	Data Integrity	Manual post-completion job part edits can cause stock drift.	Enforce job status lockouts prior to part modifications.